

## PRACTICAL 20 -

**Objective** - Write a program in c to delete duplicate elements from an array of 20 integers.

### CODE -

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[20];
5     int n = 20;
6     int i, j, k;
7     // Input array elements
8     printf("Enter %d integer elements:\n", n);
9     for (i = 0; i < n; i++) {
10         scanf("%d", &arr[i]);
11     }
12     // Delete duplicate elements
13     for (i = 0; i < n; i++) {
14         for (j = i + 1; j < n; j++) {
15             if (arr[i] == arr[j]) {
16                 // Shift elements to remove the duplicate
17                 for (k = j; k < n - 1; k++) {
18                     arr[k] = arr[k + 1];
19                 }
20                 n--; // Decrease the size of the array
21                 j--; // Adjust j to re-check the current position
22             }
23         }
24     }
25     // Print the array after deleting duplicates
26     printf("Array after deleting duplicates:\n");
27     for (i = 0; i < n; i++) {
28         printf("%d ", arr[i]);
29     }
30     printf("\n");
31     return 0;
32 }
```

### OUTPUT -

```
• Enter 20 integer elements:
  3 4 5 6 3 5 7 8 56 53 1 23 12 24 23 34 6 53 13 7
  Array after deleting duplicates:
  3 4 5 6 7 8 56 53 1 23 12 24 34 13
```